

확률·랜덤프로세스 기말대비 기말 직전 최종복습 — solved 20選 (L05~L12 전범위 · 풀이/핵심 포함)

① 한 문제당 1페이지 · 상단=문제, 하단=핵심/답 ② 1시간 전 회독용: 답 가리고 접근만 떠올려보기 ③ (★주의) L09 CLT·표본평균은 solved에 없음 → 노트로 별도 점검

떠올릴 개념

- 연속RV: $f(x)=dF/dx$, $F(x)=P[X\leq x]$; 모든 PMF/PDF는 구간 밖 = 0 (항상 명시)
- 균등 $U(a,b)$: $f=1/(b-a)$, $E=(a+b)/2$, $Var=(b-a)^2/12$ / 지수 $Exp(\lambda)$: $f=\lambda e^{-\lambda x}(x\geq 0)$, $E=1/\lambda$, $Var=1/\lambda^2$, 무기역성
- 정규: $Z=(X-\mu)/\sigma$ 표준화 $\rightarrow P=\Phi$ 로; $\Phi(-z)=1-\Phi(z)$
- 결합: 주변=합/적분, 조건부 $f_{X|Y}=f_{X,Y}/f_Y$, 의존영역은 적분한계 주의(삼각/원판)
- 공분산 $Cov=E[XY]-E[X]E[Y]$, $\rho=Cov/(\sigma_X \sigma_Y)\in[-1,1]$, $Var(X\pm Y)=VarX+VarY\pm 2Cov$
- 반복기댓값 $E[X]=E[E[X|Y]]$ ($E[X|Y]$ 는 RV!), 전분산 $Var(X)=E[Var(X|Y)]+Var(E[X|Y])$
- 결합정규: 선형결합도 정규, $\rho=0 \iff$ 독립(이 분포 한정), MMSE가 선형이 됨
- 추정: $MMSE=E[X|Y]$; 선형추정 $a^*=Cov/Var(Y)$, $b^*=E[X]-a^*E[Y]$, $e^*_L=VarX-Cov^2/VarY$, 직교원리 $E[(X-\hat{X})Y]=0$

문항 목차 (총 20문항)

- [1] L05 · PSP 4.3.2 — CDF $\rightarrow$ PDF 미분(균등) (기본)
- [2] L05 · PSP 4.5.13 — 지수분포 식별·CDF·E·Var (기본)
- [3] L05 · PSP 4.5.10 — 균등 $E[g(X)]$ ·홀함수=0 (기본)
- [4] L05 · PSP 4.4.6 — CDF $\rightarrow$ PDF·모멘트(치환) (중)
- [5] L06 · PSP 4.6.1 — 정규 표준화 Φ 확률 (기본)
- [6] L06 · PSP 5.2.1 — 결합PMF 정규화·부등식확률 (기본)
- [7] L06 · PSP 5.5.1 — 삼각영역 균등 결합PDF·주변 (중)
- [8] L07 · PSP 4.4.1 — $Y=X^2$ 변환·E·Var·Jensen (중)
- [9] L07 · PSP 5.7.5 — ρ 로 $Var(X+Y)$ 역산 (기본)
- [10] L07 · PSP 5.8.1 — $Y=X+Z$ 상관계수·독립판정 (기본)
- [11] L08 · PSP 7.5.2 — 결합PDF 조건부기댓값 $E[Y|X]$ (기본)
- [12] L08 · PSP 7.5.4 — 반복기댓값 $E[E[B|A]]$ (중)
- [13] L08 · PSP 7.5.5 — 포아송 결합·반복기댓값 (중)
- [14] L08 · PSP 7.6.3 — 혼합집단 전분산·Simpson (상)
- [15] L08 · IPSRP 5.4.36 — 결합정규 선형결합·공분산 (중)
- [16] L11 · PSP 12.1.1 — 사건조건 MMSE·무기역성 (기본)
- [17] L11 · PSP 12.1.4 — MMSE=조건부평균 $E[X|Y]$ (중)
- [18] L12 · PSP 12.1.6 — 결합가우시안 MMSE=선형 (중)
- [19] L12 · PSP 12.2.2 — 신호+독립잡음 선형추정 (기본)
- [20] L12 · PSP 12.2.8 — 원점통과 aY·직교원리 (중)

4.3.2 ● The cumulative distribution function of random variable X is

$$F_X(x) = \begin{cases} 0 & x < -1, \\ (x+1)/2 & -1 \leq x < 1, \\ 1 & x \geq 1. \end{cases}$$

Find the PDF $f_X(x)$ of X .

■ 핵심 / 접근

$f_X(x) = dF_X/dx$, 구간별 미분.

선형 CDF → 기울기 = 일정한 PDF(균등).

구간 밖 0 명시.

답 $f_X(x) = 1/2$ ($-1 \leq x < 1$), 그 밖 0 (Uniform(-1,1))

4.5.13 ■ The probability density function of random variable X is

$$f_X(x) = \begin{cases} (1/2)e^{-x/2} & x \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What is $P[1 \leq X \leq 2]$?
- (b) What is $F_X(x)$, the cumulative distribution function of X ?
- (c) What is $E[X]$, the expected value of X ?
- (d) What is $\text{Var}[X]$, the variance of X ?

■ 핵심 / 접근

$\lambda e^{(-\lambda x)}$ 에서 $\lambda=1/2$ 식별. 구간 밖 0.

CDF= $1-e^{(-\lambda x)}$, $E=1/\lambda$, $\text{Var}=1/\lambda^2$.

지수분포: $\text{Var}=(E[X])^2$

답 $P \approx 0.239$, $F = 1 - e^{(-x/2)}$, $E[X] = 2$, $\text{Var} = 4$

4.5.10 ■ X is a continuous uniform $(-5, 5)$ random variable.

- (a) What is the PDF $f_X(x)$?
- (b) What is the CDF $F_X(x)$?
- (c) What is $E[X]$?
- (d) What is $E[X^5]$?
- (e) What is $E[e^X]$?

■ 핵심 / 접근

$f_X = 1/(b-a)$, 구간 밖 0. $E[X] = (a+b)/2$.

$E[g(X)] = \int g f dx$. 홀함수+원점대칭 $\rightarrow 0$.

$E[e^x] = (e^5 - e^{-5})/10 \approx 14.84$

답 $f = 1/10$ ($-5 \leq x \leq 5$), $E[X] = 0$, $E[X^5] = 0$, $E[e^x] \approx 14.84$

4.4.6 ■ The cumulative distribution function of random variable V is

$$F_V(v) = \begin{cases} 0 & v < -5, \\ (v+5)^2/144 & -5 \leq v < 7, \\ 1 & v \geq 7. \end{cases}$$

(a) What are $E[V]$ and $\text{Var}[V]$?

(b) What is $E[V^3]$?

■ 핵심 / 접근

미분: $f_V(v) = (v+5)/72$, 구간 밖 0.

$u = v+5$ 치환 $\rightarrow E[U^k] = 12^{k+2}/(72(k+2))$.

$V = U - 5$ 선형변환으로 모멘트 환산.

답 $E[V]=3$, $\text{Var}[V]=8$, $E[V^3]=431/5=86.2$

4.6.1 ● The peak temperature T , as measured in degrees Fahrenheit, on a July day in New Jersey is the Gaussian $(85, 10)$ random variable. What is $P[T > 100]$, $P[T < 60]$, and $P[70 \leq T \leq 100]$?

■ 핵심 / 접근

Yates는 $N(\mu, \sigma)$: $\sigma=10$. $Z=(T-\mu)/\sigma$.

$P[Z > z] = 1 - \Phi(z)$, $\Phi(-z) = 1 - \Phi(z)$.

경계를 σ 단위로 환산이 핵심.

답 0.0668, 0.0062, 0.8664

5.2.1 ● Random variables X and Y have the joint PMF

$$P_{X,Y}(x,y) = \begin{cases} cxy & x = 1, 2, 4; \quad y = 1, 3, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What is the value of the constant c ?
- (b) What is $P[Y < X]$?
- (c) What is $P[Y > X]$?
- (d) What is $P[Y = X]$?
- (e) What is $P[Y = 3]$?

■ 핵심 / 접근

가분형: $\sum \sum xy = (\sum x)(\sum y) = 28 \rightarrow c = 1/28$. 표 깔고 조건 칸만 합산. 구간 밖 0.
주변 $P_Y(y) = y/4$ 로 검산.

답 $c = 1/28$; $P[Y < X] = 9/14$; $P[Y > X] = 9/28$; $P[Y = X] = 1/28$; $P[Y = 3] = 3/4$

5.5.1 ● Random variables X and Y have the joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 1/2 & -1 \leq x \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

Sketch the region of nonzero probability and answer the following questions.

- (a) What is $P[X > 0]$?
- (b) What is $f_X(x)$?
- (c) What is $E[X]$?

■ 핵심 / 접근

x 고정 $\rightarrow y$ 는 x 부터 1까지(의존 한계!). $f_X(x) = \frac{1}{2}(1-x)$, $-1 \leq x \leq 1$ 밖 0.
 균등=확률은 넓이비. $E[X] = \int x \cdot f_X dx$.

답 $f_X(x) = \frac{1}{2}(1-x)$; $P[X > 0] = 1/4$; $E[X] = -1/3$

4.4.1 ● Random variable X has PDF

$$f_X(x) = \begin{cases} 1/4 & -1 \leq x \leq 3, \\ 0 & \text{otherwise.} \end{cases}$$

Define the random variable Y by $Y = h(X) = X^2$.

- (a) Find $E[X]$ and $\text{Var}[X]$.
- (b) Find $h(E[X])$ and $E[h(X)]$.
- (c) Find $E[Y]$ and $\text{Var}[Y]$.

■ 핵심 / 접근

$E[g(X)] = \int g \cdot f \, dx$ 직접적분($E[X], E[X^2], E[X^4]$).

$\text{Var}[Y] = E[X^4] - (E[X^2])^2$.

Jensen: 볼록 $x^2 \rightarrow h(E[X]) \leq E[h(X)]$.

답 $E[X]=1, \text{Var}[X]=4/3; h(E[X])=1, E[h(X)]=7/3; E[Y]=7/3, \text{Var}[Y]=304/45 \approx 6.76$

5.7.5 ● X and Y are random variables with $E[X] = E[Y] = 0$ and $\text{Var}[X] = 1$, $\text{Var}[Y] = 4$ and correlation coefficient $\rho = 1/2$. Find $\text{Var}[X + Y]$.

■ 핵심 / 접근

$\text{Cov} = \rho \cdot \sqrt{(\text{Var}X \cdot \text{Var}Y)} = \frac{1}{2} \cdot 2 = 1$. $\text{Var}[X+Y] = \text{Var}X + \text{Var}Y + 2\text{Cov}$.

평균0은 미끼(분산은 평행이동 불변).

답 $\text{Var}[X+Y] = 1 + 4 + 2 = 7$

5.8.1 ● X and Z are independent random variables with $E[X] = E[Z] = 0$ and variance $\text{Var}[X] = 1$ and $\text{Var}[Z] = 16$. Let $Y = X + Z$. Find the correlation coefficient ρ of X and Y . Are X and Y independent?

■ 핵심 / 접근

$\text{Cov}(X, X+Z) = \text{Var}X + \text{Cov}(X, Z) = 1 + 0 = 1$. $\text{Var}Y = 1 + 16 = 17$.

$\rho = \text{Cov} / \sqrt{(\text{Var}X \cdot \text{Var}Y)} = 1 / \sqrt{17}$. $\rho \neq 0 \rightarrow$ 독립 아님.

답 $\rho = 1/\sqrt{17} \approx 0.243$; X, Y 독립 아님

7.5.2● Let random variables X and Y have joint PDF $f_{X,Y}(x,y)$ given in Problem 7.5.1. Find the PDF $f_X(x)$, the conditional PDF $f_{Y|X}(y|x)$, and the conditional expected value $E[Y|X = x]$.

■ 핵심 / 접근

x 고정 $\rightarrow Y \sim U[0,x]$. $f_X(x)=2x$ (밖0), $f_{Y|X}=1/x$.
 $E[Y|X=x]=x/2$ (구간 중점).

답 $f_X=2x$, $f_{Y|X}=1/x$, $E[Y|X=x]=x/2$

7.5.4 ■ For random variables A and B given in Problem 7.5.3, let $U = E[B|A]$. Find the PMF $P_U(u)$. What is $E[U] = E[E[B|A]]$?

■ 핵심 / 접근

함정: $E[B|A]$ 는 숫자 아닌 RV $g(A)$. U 의 PMF는 A 확률을 $g(a)$ 로 옮김.
 $E[U] = E[E[B|A]] = E[B]$ (반복기댓값).

답 $P_U(1/2) = 2/3$, $P_U(2/3) = 1/3$, $E[U] = E[B] = 5/9$

7.5.5 ■ Random variables N and K have the joint PMF

$$P_{N,K}(n, k) = \begin{cases} \frac{100^n e^{-100}}{(n+1)!} & n=0,1,\dots; \\ & k=0,1,\dots,n, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) Find the marginal PMF $P_N(n)$ and the conditional PMF $P_{K|N}(k|n)$.
- (b) Find the conditional expected value $E[K|N = n]$.
- (c) Express the random variable $E[K|N]$ as a function of N and use the iterated expectation to find $E[K]$.

■ 핵심 / 접근

결합이 k 에 무관 $\Rightarrow K|N \sim \text{Uniform}\{0..n\}$, $P_{\{K|N\}} = 1/(n+1)$ (밖 0).

$N \sim \text{Poisson}(100)$, $E[K|N] = N/2$, $E[K] = E[N]/2$.

답 $P_N \sim \text{Poi}(100)$, $E[K|N=n] = n/2$, $E[K] = 50$

7.6.3 ■ A study of bicycle riders found that a male cyclist's speed X (in miles per hour over a 100-mile "century" ride) and weight Y (kg) could be modeled by a bivariate Gaussian PDF $f_{X,Y}(x,y)$ with parameters $\mu_X = 20$, $\sigma_X = 2$, $\mu_Y = 75$, $\sigma_Y = 5$ and $\rho_{X,Y} = -0.6$. In addition, a female cyclist's speed X' and weight Y' could be modeled by a bivariate Gaussian PDF $f_{X',Y'}(x',y')$ with parameters $\mu_{X'} = 15$, $\sigma_{X'} = 2$, $\mu_{Y'} = 50$, $\sigma_{Y'} = 5$ and $\rho_{X',Y'} = -0.6$. For men and women, the negative correlation of speed and weight reflects the common wisdom that fast cyclists are thin. As it happens, cycling is much more popular among men than women; in a mixed group of cyclists, a cyclist is a male with probability $p = 0.80$.

You suspect it's OK to ignore the differences between men and women since for both groups, weight and speed are negatively correlated, with $\rho = -0.6$. To convince yourself this is OK, you decide to study the speed $\tilde{X}$ and weight $\tilde{Y}$ of a cyclist randomly chosen from a large mixed group of male and female cyclists. How are $\tilde{X}$ and $\tilde{Y}$ correlated? Explain your answer.

■ 핵심 / 접근

전공분산: $\text{Cov} = E[\text{그룹내Cov}] + \text{그룹평균 간 Cov}$.

전분산: $\text{Var} = E[\text{Var}|M] + \text{Var}(E[\cdot|M])$.

그룹내 음(-)이라도 평균차가 압도→전체 양(+), 성별 무시 금지.

답 Cov=+14, Var(X)=8, Var(Y)=125, $\rho \approx +0.44$

Problem 36

Let X and Y be jointly normal random variables with parameters $\mu_X = -1$, $\sigma_X^2 = 4$, $\mu_Y = 1$, $\sigma_Y^2 = 1$, and $\rho = -\frac{1}{2}$.

- Find $P(X + 2Y \leq 3)$.
- Find $Cov(X - Y, X + 2Y)$.

■ 핵심 / 접근

결합정규 선형결합=정규 → 평균·분산만 구해 Φ .

$Var(aX+bY)=a^2VarX+b^2VarY+2ab\cdot Cov$, $Cov=\rho\sigma_X\sigma_Y$.

공분산 쌍선형 전개.

답 (a) $\Phi(1)\approx 0.8413$ (b) 1

12.1.1 ● Generalizing the solution of Example 12.2, let the call duration T be an exponential (λ) random variable. For $t_0 > 0$, show that the minimum mean square error estimate of T , given that $T > t_0$ is

$$\hat{T} = t_0 + E[T].$$

■ 핵심 / 접근

MMSE=조건부평균 $E[T|T>t_0]$.

지수 무기역성: 잔여시간 $\sim \text{Exp}(\lambda)$, $E[R]=E[T]=1/\lambda$.

f 는 $t \geq 0$ 밖 0; 조건부 f 는 $t > t_0$ 밖 0.

답 $\hat{T} = t_0 + E[T] = t_0 + 1/\lambda$

12.1.4 ● X and Y have the joint PDF

$$f_{X,Y}(x,y) = \begin{cases} 6(y-x) & 0 \leq x \leq y \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

- (a) What is $f_{X|Y}(x|y)$?
- (b) What is $\hat{x}_M(y)$, the MMSE estimate of X given $Y = y$?
- (c) What is $f_{Y|X}(y|x)$?
- (d) What is $\hat{y}_M(x)$, the MMSE estimate of Y given $X = x$?

■ 핵심 / 접근

MMSE=조건부평균 $E[X|Y]$; 조건부밀도=결합÷주변(구간 밖 0).
 $f_Y=3y^2$, $f_X=3(1-x)^2$; 선형삼각분포 평균은 긴 변 쪽 1/3 지점.

답 $\hat{x}_M(y)=y/3$, $\hat{y}_M(x)=(x+2)/3$; $f_{X|Y}=2(y-x)/y^2$

12.1.6 ■ A signal X and noise Z are independent Gaussian $(0, 1)$ random variables, and $Y = X + Z$ is a noisy observation of the signal X . Usually, we want to use Y to estimate of X ; however, in this problem we will use Y to estimate the noise Z .

- (a) Find $\hat{Z}(Y)$, the MMSE estimator of Z given Y .
- (b) Find the mean squared error $e = E[(Z - \hat{Z}(Y))^2]$.

■ 핵심 / 접근

결합가우시안 → MMSE가 선형=LLMS: $\hat{Z} = E[Z] + (\text{Cov}/\text{Var}Y)(Y - E[Y])$.

$\text{Cov}(Z, Y) = \text{Var}Z = 1$, $\text{Var}Y = 2$; $\text{MSE} = (1 - \rho^2)\text{Var}Z$, $\rho^2 = 1/2$.

답 $\hat{Z}(Y) = Y/2$, $e = 1/2$

12.2.2 ● A telemetry voltage V , transmitted from a position sensor on a ship's rudder, is a random variable with PDF

$$f_V(v) = \begin{cases} 1/12 & -6 \leq v \leq 6, \\ 0 & \text{otherwise.} \end{cases}$$

A receiver in the ship's control room receives $R = V + X$. The random variable X is a Gaussian $(0, \sqrt{3})$ noise voltage that is independent of V . The receiver uses R to calculate a linear estimate of the telemetry voltage: $\hat{V} = aR + b$. Find

- the expected received voltage $E[R]$,
- the variance $\text{Var}[R]$ of the received voltage,
- the covariance $\text{Cov}[V, R]$ of the transmitted and received voltages,
- a^* and b^* , the optimum coefficients in the linear estimate,
- e_L^* , the minimum mean square error of the estimate.

■ 핵심 / 접근

독립잡음: $\text{Var}[R] = \text{Var}V + \text{Var}X$, $\text{Cov}[V, R] = \text{Var}V$ (잡음 소거).

$a^* = \text{Cov}/\text{Var}R < 1$ (수축), $e_L^* = \text{Var}V - \text{Cov}^2/\text{Var}R$; $\text{VarUnif} = (\beta - \alpha)^2/12$.

답 $E[R]=0$, $\text{Var}[R]=15$, $\text{Cov}=12$, $a^*=4/5$, $b^*=0$, $e_L^*=12/5=2.4$

12.2.8 ■ For random variables X and Y , we wish to use Y to estimate X . However, our estimate must be of the form $\hat{X} = aY$.

- (a) Find a^* , the value of a that minimizes the mean square error $e = E[(X - aY)^2]$.
- (b) For $a = a^*$, what is the minimum mean square error e^* ?
- (c) Under what conditions is $\hat{X}$ the LMSE estimate of X ?

■ 핵심 / 접근

$a^* = E[XY]/E[Y^2]$; 오차 $X - a^*Y$ 가 Y 와 직교: $E[(X - a^*Y)Y] = 0$.

$e^* = E[X^2] - (E[XY])^2/E[Y^2]$.

aY 가 LMSE $\Leftrightarrow$ 아핀 상수항 $b_L = 0$ (특히 $E[X] = E[Y] = 0$).

답 $a^* = E[XY]/E[Y^2]$, $e^* = E[X^2] - (E[XY])^2/E[Y^2]$